

$\sigma_{2+}^{\#1} + \alpha\beta$	$\sigma_{2+}^{\#1} + \alpha\beta$	$\tau_{2+}^{\#1} + \alpha\beta$	$\sigma_{2-}^{\#1} + \alpha\beta\chi$
0	0	0	0
$\tau_{2+}^{\#1} + \alpha\beta$	0	0	0
$\sigma_{2-}^{\#1} + \alpha\beta\chi$	0	0	0

$\mathcal{A}_{2+}^{\#1} + \alpha\beta$	$\mathcal{A}_{2+}^{\#1} + \alpha\beta$	$\mathcal{A}_{2-}^{\#1} + \alpha\beta\chi$
0	0	0
$f_{2+}^{\#1} + \alpha\beta$	0	0
$\mathcal{A}_{2-}^{\#1} + \alpha\beta\chi$	0	0

$\mathcal{A}_{0+}^{\#1} +$	$\mathcal{A}_{0+}^{\#1} +$	$f_{0+}^{\#1} +$	$f_{0+}^{\#2} +$	$\mathcal{A}_{0-}^{\#1} + \alpha\beta\chi$
$\frac{3\beta_3}{2}$	$-\frac{3i\beta_3k}{\sqrt{2}}$	0	0	0
$f_{0+}^{\#1} +$	$\frac{3i\beta_3k}{\sqrt{2}}$	0	0	0
$f_{0+}^{\#2} +$	0	0	0	0
$\mathcal{A}_{0-}^{\#1} + \alpha\beta\chi$	0	0	0	0

Source constraints	#
$\sigma_{0-}^{\#1\alpha\beta\chi} == 0$	1
$\tau_{0+}^{\#2} == 0$	1
$\tau_{0+}^{\#1} - 2\,i\,k\,\sigma_{0+}^{\#1} == 0$	1
$\tau_{1-}^{\#2\alpha} + 2\,i\,k\,\sigma_{1-}^{\#2\alpha} == 0$	3
$\tau_{1-}^{\#1\alpha} == 0$	3
$\tau_{1+}^{\#1\alpha\beta} == 0$	3
$\sigma_{1+}^{\#2\alpha\beta} == 0$	3
$\sigma_{1+}^{\#1\alpha\beta} == 0$	3
$\sigma_{2-}^{\#1\alpha\beta\chi} == 0$	5
$\tau_{2+}^{\#1\alpha\beta} == 0$	5
$\sigma_{2+}^{\#1\alpha\beta} == 0$	5
Total #:	33

	$\sigma_{0+}^{\#1}$	$\tau_{0+}^{\#1}$	$\tau_{0+}^{\#2}$	$\sigma_{0-}^{\#1\alpha\beta\chi}$
$\sigma_{0+}^{\#1} +$	$\frac{1}{(1+2\,k^2)\,\text{Det}\,(0^+)}$	$-\frac{i\,\sqrt{2}\,k}{(1+2\,k^2)\,\text{Det}\,(0^+)}$	0	0
$\tau_{0+}^{\#1} +$	$\frac{i\,\sqrt{2}\,k}{(1+2\,k^2)\,\text{Det}\,(0^+)}$	$\frac{1}{(1+\frac{1}{2\,k^2})\,\text{Det}\,(0^+)}$	0	0
$\tau_{0+}^{\#2} +$	0	0	0	0
$\sigma_{0-}^{\#1} + \alpha\beta\chi$	0	0	0	0

Lagrangian density

$$\begin{aligned}
 & -\beta_3\,\mathcal{A}^{\prime\theta}{}_{\mu}{}_{}\mathcal{A}^{\mu}{}_{}{}_{}\partial_{\theta}f^{\prime\theta}-\beta_3\,\mathcal{A}^{\mu\nu}{}_{\mu}\partial_{\theta}f^{\prime\theta}- \\
 & \beta_3\,\mathcal{A}^{\mu}{}_{}{}_{}\partial^{\theta}f^{\prime}{}_{}{}_{}+\beta_3\,\mathcal{A}^{\mu}{}_{}{}_{}\partial^{\theta}f^{\prime}{}_{}{}_{}+\beta_3\partial_{\theta}f^{\mu}{}_{}\partial^{\theta}f^{\prime}{}_{}{}_{}+ \\
 & \frac{1}{6}\chi\partial_{\theta}\mathcal{A}^{\nu}{}_{}{}_{}\partial^{\mu}\mathcal{A}^{\prime\theta}{}_{}{}_{}-\frac{1}{6}\chi\partial_{\mu}\mathcal{A}^{\nu}{}_{}{}_{}\partial^{\mu}\mathcal{A}^{\prime\theta}{}_{}{}_{}-\frac{1}{6}\chi\partial_{\alpha}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial_{\theta}f^{\prime\theta}+ \\
 & \frac{1}{6}\chi\partial_{\alpha}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial_{\theta}f^{\prime\theta}-\frac{1}{6}\chi\partial_{}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial_{\theta}f^{\prime\theta}+ \\
 & \frac{1}{6}\chi\partial_{\mu}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial_{\theta}f^{\prime\theta}+\frac{1}{6}\chi\partial_{\alpha}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial^{\theta}f^{\prime}{}_{}{}_{}-\frac{1}{6}\chi\partial_{\alpha}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial^{\theta}f^{\prime}{}_{}{}_{}+ \\
 & \frac{1}{6}\chi\partial_{\theta}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial^{\theta}f^{\prime}{}_{}{}_{}-\frac{1}{6}\chi\partial_{\mu}\mathcal{A}^{}{}_{}{}_{}\partial^{\mu}\partial^{\theta}f^{\prime}{}_{}{}_{}-2\,\beta_3\partial^{\theta}f^{\prime}{}_{}{}_{}\partial_{\nu}f^{}{}_{}{}_{}^{}+ \\
 & \beta_3\partial_{\theta}f^{}{}_{}{}_{}^{}\partial_{\nu}f^{\mu\nu}+\frac{1}{6}\chi\partial^{\mu}\mathcal{A}^{\prime\theta}{}_{}{}_{}\partial_{\nu}\mathcal{A}^{}{}_{}{}_{}^{}-\frac{1}{6}\chi\partial^{\mu}\mathcal{A}^{\prime\theta}{}_{}{}_{}\partial_{\nu}\mathcal{A}^{}{}_{}{}_{}^{}
 \end{aligned}$$

Added source term(s): $f^{\alpha\beta}\,\tau_{\alpha\beta}+\mathcal{A}^{\alpha\beta\chi}\,\sigma_{\alpha\beta\chi}$

$$\text{Det}\,(1^-)\rightarrow -\frac{1}{288}\,\chi^2\,k^4\,(1+2\,k^2)$$

$$\text{Det}\,(0^+)\rightarrow \frac{3}{2}\,\beta_3\,(1+2\,k^2)$$

$\sigma_{1+}^{\#1} + \alpha\beta$	$\sigma_{1+}^{\#2} + \alpha\beta$	$\tau_{1+}^{\#1} + \alpha\beta$	$\sigma_{1-}^{\#1} + \alpha$	$\sigma_{1-}^{\#2} + \alpha$	$\tau_{1-}^{\#1} + \alpha$	$\tau_{1-}^{\#2} + \alpha$
0	0	0	0	0	0	0
$\sigma_{1+}^{\#2} + \alpha\beta$	0	0	0	0	0	0
$\tau_{1+}^{\#1} + \alpha\beta$	0	0	0	0	0	0
$\sigma_{1-}^{\#1} + \alpha$	0	0	0	0	0	0
$\sigma_{1-}^{\#2} + \alpha$	0	0	0	0	0	0
$\tau_{1-}^{\#1} + \alpha$	0	0	0	0	0	0
$\tau_{1-}^{\#2} + \alpha$	0	0	0	0	0	0
$\sigma_{1+}^{\#1} + \alpha\beta$	$\sigma_{1+}^{\#2} + \alpha\beta$	$\tau_{1+}^{\#1} + \alpha\beta$	$\sigma_{1-}^{\#1} + \alpha$	$\sigma_{1-}^{\#2} + \alpha$	$\tau_{1-}^{\#1} + \alpha$	$\tau_{1-}^{\#2} + \alpha$
0	0	0	$\frac{\beta_3\,(\frac{1}{2}+k^2)}{\text{Det}\,(1^-)}$	$\frac{12\,\beta_3+\chi\,k^2}{12\,\sqrt{2}\,\text{Det}\,(1^-)}$	0	$\frac{i\,k\,(12\,\beta_3+\chi\,k^2)}{12\,\text{Det}\,(1^-)}$
0	0	0	0	$\frac{6\,\beta_3+\chi\,k^2}{(6+12\,k^2)\,\text{Det}\,(1^-)}$	0	$\frac{i\,k\,(6\,\beta_3+\chi\,k^2)}{3\,\sqrt{2}\,(1+2\,k^2)\,\text{Det}\,(1^-)}$
0	0	0	0	0	0	0
0	0	0	$-\frac{i\,k\,(12\,\beta_3+\chi\,k^2)}{12\,\text{Det}\,(1^-)}$	$-\frac{i\,k\,(6\,\beta_3+\chi\,k^2)}{3\,\sqrt{2}\,(1+2\,k^2)\,\text{Det}\,(1^-)}$	0	$\frac{6\,\beta_3\,k^2+\chi\,k^4}{(3+6\,k^2)\,\text{Det}\,(1^-)}$

$\mathcal{A}_{1+}^{\#1} + \alpha\beta$	$\mathcal{A}_{1+}^{\#2} + \alpha\beta$	$f_{1+}^{\#1} + \alpha\beta$	$\mathcal{A}_{1-}^{\#1} + \alpha$	$\mathcal{A}_{1-}^{\#2} + \alpha$	$f_{1-}^{\#1} + \alpha$	$f_{1-}^{\#2} + \alpha$
0	0	0	0	0	0	0
$\mathcal{A}_{1+}^{\#2} + \alpha\beta$	0	0	0	0	0	0
$f_{1+}^{\#1} + \alpha\beta$	0	0	0	0	0	0
$\mathcal{A}_{1-}^{\#1} + \alpha$	0	0	0	0	0	0
$\mathcal{A}_{1-}^{\#2} + \alpha$	0	0	0	0	0	0
$f_{1-}^{\#1} + \alpha$	0	0	0	0	0	0
$f_{1-}^{\#2} + \alpha$	0	0	0	0	0	0
$\mathcal{A}_{1+}^{\#1} + \alpha\beta$	$\mathcal{A}_{1+}^{\#2} + \alpha\beta$	$f_{1+}^{\#1} + \alpha\beta$	$\mathcal{A}_{1-}^{\#1} + \alpha$	$\mathcal{A}_{1-}^{\#2} + \alpha$	$f_{1-}^{\#1} + \alpha$	$f_{1-}^{\#2} + \alpha$
0	0	0	$\beta_3+\frac{\chi\,k^2}{6}$	$-\frac{12\,\beta_3+\chi\,k^2}{12\,\sqrt{2}}$	0	$-\frac{1}{12}\,i\,(12\,\beta_3\,k+\chi\,k^3)$
0	0	0	0	$\frac{\beta_3}{2}$	0	$\frac{i\,\beta_3\,k}{\sqrt{2}}$
0	0	0	0	0	0	0
0	0	0	$-\frac{i\,\beta_3\,k}{\sqrt{2}}$	$\beta_3\,k^2$	0	$\beta_3\,k^2$

Quadratic pole

Pole residue: $\frac{1}{\beta_3} + \frac{72\,\beta_3}{\chi^2\,p^4} + \frac{12\,(24\,\beta_3+\chi)}{\chi^2\,p^2} > 0$

Polarisations: 1

Unitarity conditions

$\chi < 0 \ \&\& \ \beta_3 > 0 \ \bigg| \bigg| \ \chi > 0 \ \&\& \ \beta_3 > 0$

(No massive particles)